

ISO/SAE 21434 TARA Threat Analysis

<https://gemini.google.com/app/86f00a5ef998e2ba>

User prompt: Act as a Senior Automotive and IoT Cybersecurity Engineer conforming to the ISO/SAE 21434 standard. Perform a preliminary Threat Analysis and Risk Assessment (TARA) — specifically Threat Identification and Damage Scenario formulation — for the system architecture modeled in Ansys Medini Analyze described below. ### System Architecture Overview The system is a multi-tier environmental/indoor monitoring platform featuring dual-path edge-to-cloud telemetry: Central Processing Unit: «Controller» Indoor Unit (IUC) Sensing Subsystem (connected to IUC): Particulate matter: PM2.5, PM10 Toxic/ambient gases: CO, CO2, NO2, SO2, O3 Geolocation: «Component» GPS (bidirectional interface with IUC) Actuation Subsystem: «Actuator» Status Display (controlled directly by IUC via Display In/Out lines) Uplink Path A (Local Edge / Cellular): IUC <-> «Component» Wi-Fi <-> «Controller» Far edge device <-> «Component» 4G <-> «Item/System» Cloud server Uplink Path B (LPWAN): IUC <-> «Component» LoRaWAN <-> «Component» LoRa GW <-> «Item/System» LoRa edge server <-> «Item/System» Cloud server Backend & Consumer Tier: «Item/System» Cloud server <-> «Item/System» User (dashboard / client application) Task Scope & Objectives 1. Apply the STRIDE threat modeling methodology systematically across: - Physical/analog sensor interfaces and buses to the IUC. - Local wireless channels (Wi-Fi, LoRaWAN, Cellular 4G, GPS RF signals). - Intermediate computing nodes (Far edge device, LoRa Gateway, LoRa edge server). - Cloud backend endpoints and end-user data delivery channels. 2. Formulate clear Threat Scenarios and map each to a concrete Damage Scenario considering vehicle/system-level safety, operation, financial impact, and user privacy (SFOP). ###Output Format Provide the results in a structured Markdown table compatible with Ansys Medini Analyze TARA tables, using the following columns: | Threat ID | Target Element / Interface | STRIDE Category | Compromised Asset & Property (Confidentiality, Integrity, Availability, Authenticity) | Threat Scenario Description | Damage Scenario (Vehicle / Passenger / Operational Impact) | |---|---|---|---|---|---| Ensure deep technical coverage of: - Sensor data spoofing/tampering (especially safety-critical gases like CO and toxic pollutants) causing suppressed alerts or false status indications on the Status Display. - Wireless jamming, replay, and eavesdropping attacks on LoRaWAN, 4G, Wi-Fi, and GPS. - Compromise or lateral movement via intermediate processing nodes (Far edge device, LoRa Edge server). Note: Please remember that there must not be any threads with more than one STRIDE category! One category per thread.

Preliminary Threat Analysis and Risk Assessment (TARA)

System Item Under Analysis: Dual-Path Environmental Monitoring Platform (IUC, Edge, LPWAN, Cloud, & Consumer Tier)

Standard Compliance: ISO/SAE 21434:2021 (Road vehicles — Cybersecurity engineering)

Methodology: STRIDE-per-element Threat Identification strictly mapped to 1:1 STRIDE categories with clear Damage Scenarios evaluating **Safety (S), Operational (O), Financial (F), and Privacy (P)** impacts.

Ansys Medini Analyze Threat Identification & Damage Scenario Table

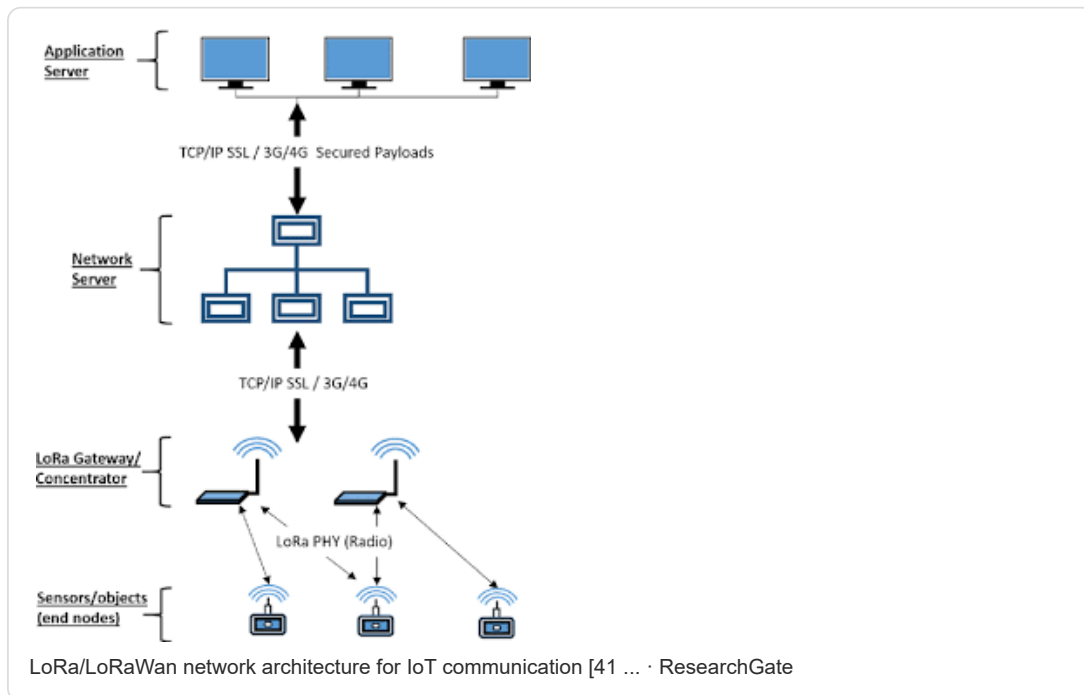
Threat ID	Target Element / Interface	STRIDE Category	Compromised Asset & Property	Threat Scenario Description	Damage Scenario (Vehicle / Passenger / Operational Impact)
THR-01	Analog/I2C/UART Sensor Bus (CO, CO2, NO2, SO2, O3)	Tampering	Gas Measurement Data (Integrity)	Attacker injects fake analog voltage or manipulates digital I2C bus signals between toxic gas sensors and IUC to clamp CO readings to near-zero levels during a real gas build-up.	Safety (High): Passengers or operators in an enclosed vehicle cabin/facility are exposed to toxic carbon monoxide without warning, leading to acute asphyxiation, severe illness, or death.

Threat ID	Target Element / Interface	STRIDE Category	Compromised Asset & Property	Threat Scenario Description	Damage Scenario (Vehicle / Passenger / Operational Impact)
THR-02	Analog/I2C Sensor Bus (PM2.5, PM10)	Spoofing	Sensor Node Identity (Authenticity)	Attacker spoof-emulates a legitimate particle sensor over the internal bus, injecting false baseline telemetry into the IUC.	Operational (Low): Ventilation/HVAC filtration triggers prematurely or fails to trigger, accelerating filter wear or degrading interior air quality.
THR-03	Display In/Out Lines («Actuator» Status Display)	Tampering	Display Command Payload (Integrity)	Attacker intercepts and modifies display driver commands sent from IUC, displaying "SAFE AIR QUALITY" when internal concentration thresholds are breached.	Safety (Medium): Occupants rely on visually false "OK" status indications, delaying cabin evacuation or manual HVAC intervention during air contamination events.
THR-04	GPS Antenna / RF Receiver Interface	Spoofing	GPS Satellite Signals (Authenticity)	Attacker transmits crafted GPS RF signals (SDR spoofing) to override true satellite signals received by the «Component» GPS.	Operational (Medium) / Safety (Low): Environmental telemetry is tagged with incorrect coordinates, compromising emergency response accuracy or automated geolocation-triggered safety policies.
THR-05	GPS RF Spectrum (1575.42 MHz L1)	Denial of Service	GPS Signal Availability (Availability)	Attacker deploys a localized RF jammer targeting the GPS L1 frequency, causing loss of position lock on the IUC.	Operational (Low): Inability to correlate sensor telemetry with location coordinates; system degrades to un-geotagged logging.
THR-06	Wi-Fi Interface (IUC <-> Far edge device)	Information Disclosure	Local Telemetry Payload (Confidentiality)	Attacker eavesdrops on unencrypted local Wi-Fi frames carrying detailed environmental data and user occupancy indicators between IUC and Far edge device.	Privacy (Medium): Threat actor tracks occupant presence and operational schedules by analyzing continuous local telemetry streams.
THR-07	Wi-Fi Channel (2.4 / 5 GHz)	Denial of Service	Local Wireless Channel (Availability)	Attacker executes a Wi-Fi deauthentication or continuous RF noise flooding attack, severing Link Path A	Operational (Medium): Edge telemetry upload via Path A fails; system must fail over to LPWAN Path B,

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				between IUC and Far edge device.	reducing telemetry resolution/bandwidth.
THR-08	LoRaWAN RF Interface (868/915 MHz)	Repudiation	Frame Sequence / Counter (Repudiation)	Attacker intercepts valid LoRaWAN packets and injects modified message frames without nonces, causing IUC or LoRa GW to reject legitimate frame history.	Operational (Low): System logs become untrusted for audit compliance; dispute over missing critical event triggers.
THR-09	LoRaWAN RF Link (IUC <-> LoRa GW)	Tampering	Uplink Sensor Data Frame (Integrity)	Attacker performs a bit-flipping / relay attack on weak LoRaWAN configurations (e.g., hardcoded AppSKey/NwkSKey), modifying payload gas values in transit.	Operational (Medium): Backend dashboard receives corrupted gas readings, generating false environmental hazard alarms and unnecessary emergency dispatch costs.
THR-10	4G Cellular Interface (Far edge device <-> Cloud)	Elevation of Privilege	Edge Router Execution Context (Elevation of Privilege)	Attacker exploits an unpatched command injection vulnerability in the 4G module driver on the Far edge device to gain root shell access.	Operational (High) / Financial (High): Far edge device is compromised and converted into a botnet node; attacker pivoting laterally into the core cloud server infrastructure.
THR-11	4G / Cellular Link	Denial of Service	Cellular Uplink Path (Availability)	Attacker uses a localized IMSI-catcher or cellular band jammer to block 4G LTE communication from the Far edge device.	Operational (Low): Complete loss of primary high-throughput Telemetry Path A; reliance shifted entirely to Path B.
THR-12	«Controller» Far edge device	Tampering	Edge Processing Logic (Integrity)	Attacker leverages local physical access (UART/JTAG debug header) on the Far edge device to alter local data aggregation algorithms.	Operational (Medium): Aggregate pollution averages sent to the Cloud server are systematically altered, obscuring localized toxic spikes.
THR-13	«Controller» Far edge device	Information Disclosure	Stored Network Keys / Credentials (Confidentiality)	Attacker extracts flash memory contents from the Far edge device, exposing cloud API keys and Wi-Fi WPA3 PSKs.	Financial (Medium) / Security Risk: Attacker uses stolen cloud API tokens to impersonate edge units and inject false system-wide metrics.

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THR-14	«Item/System» LoRa edge server	Elevation of Privilege	Server Process Privilege (Elevation of Privilege)	Attacker exploits a buffer overflow in the LoRaWAN MAC network server daemon to escalate privileges to root.	Operational (High): Attacker controls the entire LoRa network tier, capable of dropping, altering, or falsifying LPWAN telemetry for all connected edge units.
THR-15	«Item/System» LoRa edge server	Spoofing	Network Server Identity (Authenticity)	Attacker deploys a rogue LoRa Edge Server, forcing gateways to terminate connections to the malicious endpoint via DNS spoofing.	Operational (Medium): Telemetry Path B is redirected away from the genuine backend, resulting in complete data loss on Path B.
THR-16	«Item/System» Cloud server	Spoofing	Cloud API Endpoint (Authenticity)	Attacker spoofs the northbound API endpoint using a forged SSL certificate, tricking Far edge devices into pushing data to a rogue server.	Privacy (High) / Financial (Medium): Interception of all fleet-wide environmental and location telemetry; exposure of proprietary deployment data.
THR-17	«Item/System» Cloud server	Information Disclosure	Central Database (Confidentiality)	Attacker executes a SQL Injection (SQLi) against the backend ingestion pipeline, dumping historical geotagged air quality records.	Privacy (High): Breach of personal location history and occupancy patterns for thousands of connected vehicles/users, violating regulatory privacy rules (GDPR).
THR-18	«Item/System» Cloud server	Denial of Service	Cloud Dashboard Service (Availability)	Attacker launches an HTTP/MQTT Distributed Denial of Service (DDoS) attack against the cloud ingestion endpoint, exhausting server resources.	Operational (High): Real-time monitoring dashboards freeze; operators and end-users lose all visibility into remote cabin/site safety metrics.
THR-19	User Application / Dashboard	Tampering	Web/App Data Rendering Layer (Integrity)	Attacker executes a Cross-Site Scripting (XSS) attack on the dashboard web app, manipulating DOM elements to suppress toxic alert banners.	Safety (Medium) / Operational (Medium): Facility managers fail to see real-time toxic gas warning popups, delaying evacuation procedures.

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THR-20	User Application / Dashboard	Elevation of Privilege	User Session Role (Elevation of Privilege)	Attacker exploits Broken Object Level Authorization (BOLA) in the mobile client API to elevate standard viewer rights to system administrator.	Operational (High) / Financial (High): Attacker remotely alters global IUC configuration thresholds, disables safety alarms, or deletes historical logs across the system fleet.



Response: Here is the breakdown of the 20 threat scenarios mapped across the **STRIDE** categories:

STRIDE Category	Category Description	Threat IDs	Count
S - Spoofing	Identity spoofing / Impersonation	THR-02, THR-04, THR-15, THR-16	4
T - Tampering	Data or hardware manipulation	THR-01, THR-03, THR-09, THR-12, THR-19	5
R - Repudiation	Actions that cannot be proven/audited	THR-08	1
I - Information Disclosure	Data leakage / Unencrypted exposure	THR-06, THR-13, THR-17	3
D - Denial of Service	Interruption or jamming of service/RF	THR-05, THR-07, THR-11, THR-18	4
E - Elevation of Privilege	Unauthorized privilege escalation / Root access	THR-10, THR-14, THR-20	3
Total			20
